PHASE 1 MACHINE LEARNING EVALUATION / TEAM T001

Flight Delay Prediction via Gradient Boosting and Temporal Deep Learning

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Predict arrival delay before departure

15 minutes

or more late at arrival

Can information available two hours before departure identify flights at greater delay risk?

Intended use

Advance contingency planning from schedule, route, airline and historical information.

Completed application

A historical prediction interface and offline schedule-shift analysis use the saved model pipeline.

Definition: BTS on-time performance convention; project prediction contract

Objectives and completed scope

1 Enforce information timing

Use permitted schedule fields and historical outcomes released before each decision.

2 Compare the required ML models

Logistic Regression and XGBoost, with an embedding MLP as a supplementary comparison.

3 Validate probabilities and alerts

Compare ranking, probability error, calibration choices and a development-selected threshold.

4 Demonstrate historical use

Saved model inference, a Streamlit interface and an offline schedule-shift simulator.

Schedule patterns and learned categorical embeddings support the comparison.

Implementation: src/; reproducible outputs: results/ and models/

2013 departures from three NYC airports

336,776

raw schedules · 21 columns

327,346

observed arrival outcomes

Observed target	Flights	Share
Delayed ≥ 15 min	80,100	24.47%
Less than 15 min late	247,246	75.53%

EWR · JFK · LGA
16 carriers
1 January–31 December 2013

Training and evaluation use observed arrival outcomes; the complete schedule supports demand features.

Sources: results/reports/dataset_profile.json; raw_dataset_audit.json; nycflights13 documentation

Leakage prevention at the decision point

Decision time
Scheduled departure – 2 h

→

Scheduled departure

Permitted current-flight inputs

Scheduled clocks, carrier, airports, distance, calendar and scheduled origin demand.

Prohibited realized inputs

Actual departure/arrival, departure delay, arrival delay and airborne time.

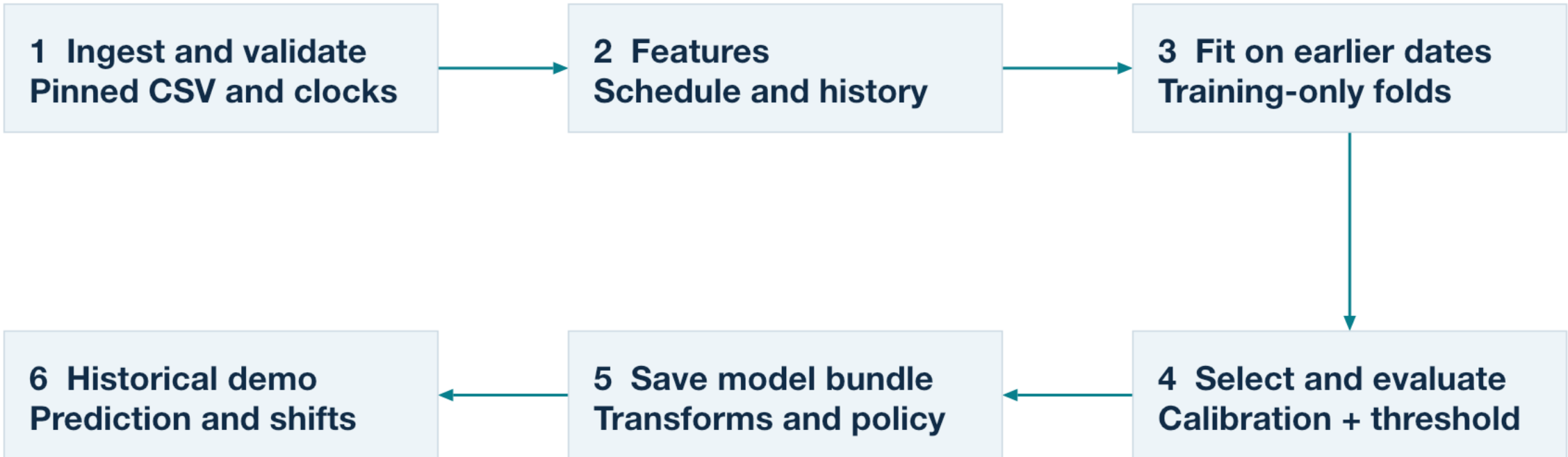
Historical outcome eligibility

Actual arrival + assumed 24 h reporting lag < local-midnight snapshot

Daily snapshots use the decision date. Automated checks verify that later outcomes cannot alter earlier features.

Contract and implementation: src/config.py, src/features.py, src/timeutils.py; tests/test_no_leakage.py

Implemented system workflow



Fixed estimator weights; historical inputs can update after outcomes are released.

Modules: data.py → features.py → experiments.py / train.py → evaluate.py → inference.py / reschedule.py

Feature engineering uses available information

Schedule and cyclic time

Scheduled departure/arrival clocks, weekday, month, weekend and distance.

Identities and scheduled demand

Carrier, origin and destination; departures in the origin hour and adjacent-hour window.

Past delay rates

Smoothed carrier, airport and route rates, plus a seven-day released-outcome window.

Model-specific selection

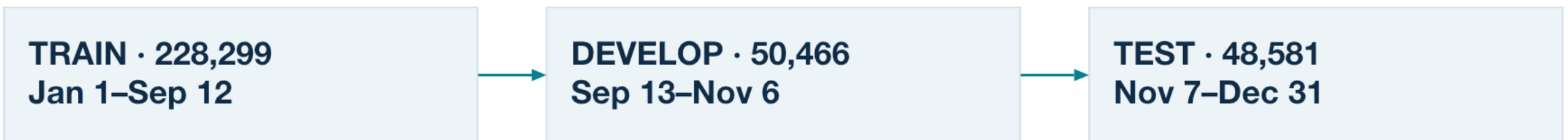
LR selects 28 numeric inputs; XGBoost selects 18. Both use carrier, origin and destination identities.

Cyclic encoding: $\sin(2\pi t / P)$, $\cos(2\pi t / P)$

Definitions: src/config.py and src/features.py; selection: results/reports/temporal_search.json

Models and chronological experimental design

Logistic Regression	XGBoost	Embedding MLP · supplementary
L2 regularization C = 0.03 · unweighted	42 trees · depth 3 learning rate 0.06	16/16/32 embeddings 128 → 64 → 32 hidden units
One-hot identities 28 numeric inputs	One-hot identities 18 numeric inputs	18 numeric inputs Adam · 15 fixed epochs



2013 date blocks · 4 expanding training folds · 227,520 labels usable for final fitting

Shared label, date partitions and information boundary; feature subsets are selected for each model.

Sources: temporal_search.json; split_manifest.json; models/pytorch_training_history.json

Metrics and the operating threshold

Ranking

Average Precision summarizes precision–recall. ROC-AUC measures ordering across score thresholds.

Probability quality

Brier loss measures probability error. Reliability curves compare predicted risk with observed frequency.

Operating rule selected using development data

XGBoost threshold: 0.206462

The displayed operating point uses the development-selected F1 rule with at least 100 alerts. Ranking and probability quality are evaluated separately.

Definitions and policy: src/evaluate.py; src/calibration.py; saved model threshold JSON files

Test results show different strengths

Model	AP ↑	ROC-AUC ↑	Brier ↓	Accuracy	Precision	Recall
LR	0.3200	0.5797	0.2396	68.17%	33.99%	19.67%
XGBoost	0.3722	0.6367	0.1910	56.79%	34.49%	67.74%
MLP supplementary	0.3936	0.6506	0.2017	47.94%	31.95%	83.04%
Constant prior	0.2686	0.5000	0.1965	73.14%	—	0.00%

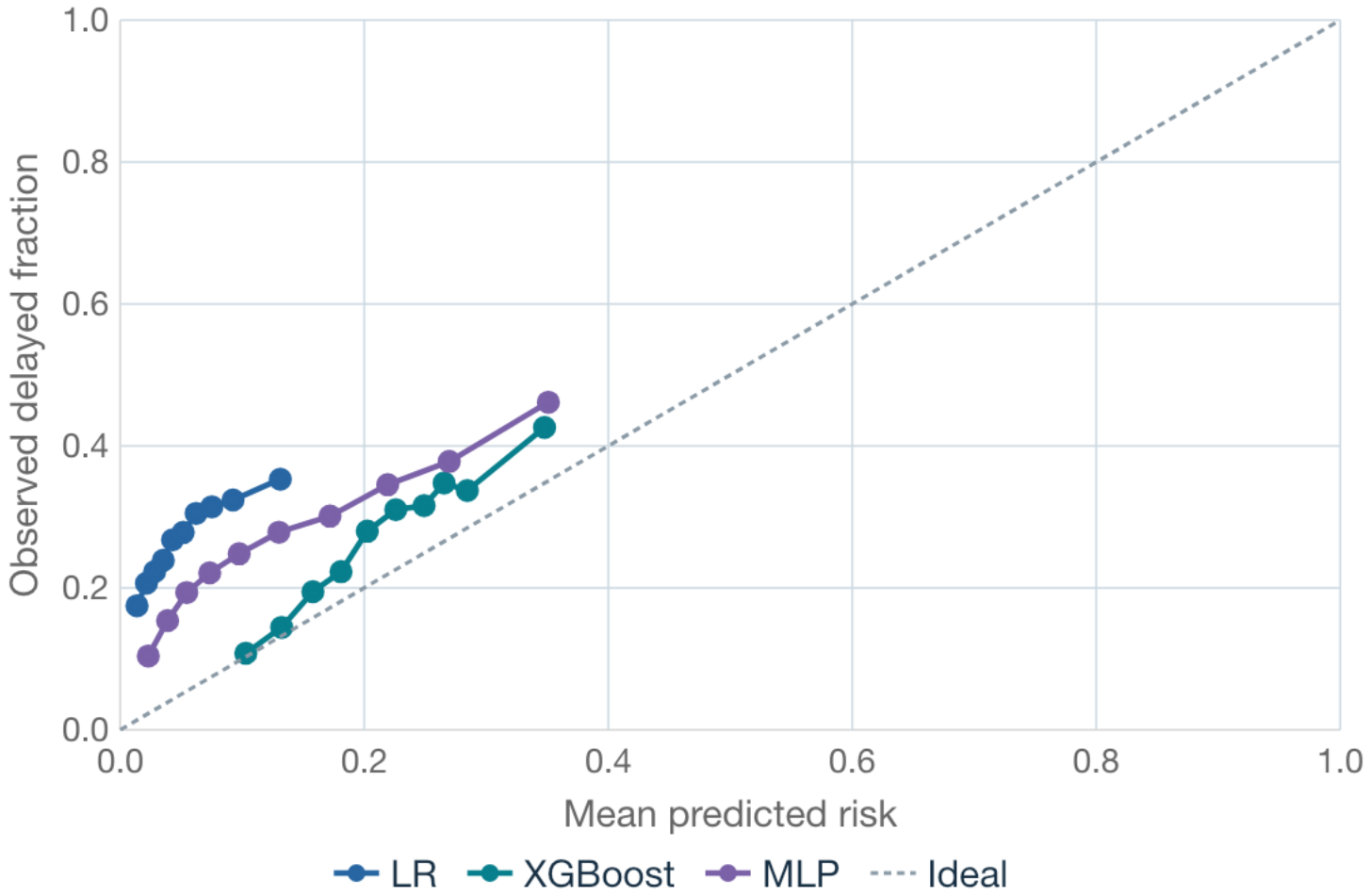
XGBoost: lowest probability error among learned models

MLP: highest observed AP (0.3936) and recall (83.04%)

Accuracy, precision and recall use development-selected F1 operating points; — denotes no alerts for the constant prior.

Retrospective test: 48,581 flights; results/metrics/evaluation_metrics.json

Probability calibration assessment



Selected treatment

Identity, selected by development Brier score, provides the saved probabilities.

Reliability assessment

Mean predicted risk is compared with observed delayed frequency across ten quantile bins.

10 quantile bins, saved test predictions; identity was selected for every estimator

Offline rescheduling changes the model score

Shift	Departure local	Predicted risk
-60 min	06:10	10.014%
-30 min	06:40	10.014%
0 min	07:10	10.281%
+30 min	07:40	10.281%
+60 min	08:10	11.648%

Selected: –30 minutes

10.281% → 10.014%
–0.267 percentage points

Equal risks favor the smaller absolute shift. Duration remains 195 minutes.

History stays fixed at the original decision time; timetable demand is recomputed against a fixed background.

Offline what-if comparison under fixed-history and fixed-background assumptions.

Saved case: AA · JFK–MIA · 7 November 2013 · original departure 07:10 local

Conclusion

Predict before departure

Schedule and released historical information support predictions two hours before departure.

Compare models on chronological data

XGBoost: lowest learned-model Brier loss, 0.1910
Embedding MLP: highest observed AP, 0.3936

Connect evaluation to a working application

Saved inference pipeline, prediction interface and offline schedule-shift analysis.